

#Étape 1 :impoter data set:

```
import os

if os.path.exists("final_training_data.jsonl"):
    os.rename("final_training_data.jsonl", "train.jsonl")
    print("importation faite avec succès!")
```

importation faite avec succès!

#Étape 2 : Préparation de l'environnement (Fix SQLite)

!pip install -U autotrain-advanced pysqlite3-binary

```
import sys
__import__('pysqlite3')
sys.modules['sqlite3'] = sys.modules.pop('pysqlite3')
print("✅ Fix SQLite appliqué !")
```

Collecting autotrain-advanced

Downloading autotrain_advanced-0.8.36-py3-none-any.whl.metadata (21 kB)

Collecting pysqlite3-binary

Downloading pysqlite3_binary-0.5.4.post2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (922 bytes)

Collecting alumentations==1.4.23 (from autotrain-advanced)

Downloading alumentations-1.4.23-py3-none-any.whl.metadata (36 kB)

Collecting datasets~=3.2.0 (from datasets[vision]~=3.2.0->autotrain-advanced)

Downloading datasets-3.2.0-py3-none-any.whl.metadata (20 kB)

Collecting evaluate==0.4.3 (from autotrain-advanced)

Downloading evaluate-0.4.3-py3-none-any.whl.metadata (9.2 kB)

Collecting ipadic==1.0.0 (from autotrain-advanced)

Downloading ipadic-1.0.0.tar.gz (13.4 MB)

13.4/13.4 MB 80.3 MB/s eta

0:00:00

etadata (setup.py) ... autotrain-advanced)

Downloading jiwer-3.0.5-py3-none-any.whl.metadata (2.7 kB)

Collecting joblib==1.4.2 (from autotrain-advanced)

Downloading joblib-1.4.2-py3-none-any.whl.metadata (5.4 kB)

Collecting loguru==0.7.3 (from autotrain-advanced)

Downloading loguru-0.7.3-py3-none-any.whl.metadata (22 kB)

Collecting pandas==2.2.3 (from autotrain-advanced)

Downloading pandas-2.2.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (89 kB)

89.9/89.9 kB 6.7 MB/s eta

0:00:00

ent already satisfied: nltk==3.9.1 in /usr/local/lib/python3.12/dist-packages (from autotrain-advanced) (3.9.1)

Collecting optuna==4.1.0 (from autotrain-advanced)

Downloading optuna-4.1.0-py3-none-any.whl.metadata (16 kB)

```

Collecting Pillow==11.0.0 (from autotrain-advanced)
  Downloading pillow-11.0.0-cp312-cp312-
manylinux_2_28_x86_64.whl.metadata (9.1 kB)
Collecting sacremoses==0.1.1 (from autotrain-advanced)
  Downloading sacremoses-0.1.1-py3-none-any.whl.metadata (8.3 kB)
Collecting scikit-learn==1.6.0 (from autotrain-advanced)
  Downloading scikit_learn-1.6.0-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (18 kB)
Collecting sentencepiece==0.2.0 (from autotrain-advanced)
  Downloading sentencepiece-0.2.0-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.7 kB)
Collecting tqdm==4.67.1 (from autotrain-advanced)
  Downloading tqdm-4.67.1-py3-none-any.whl.metadata (57 kB)


---


57.7/57.7 kB 4.6 MB/s eta
0:00:00
autotrain-advanced)
  Downloading werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Collecting xgboost==2.1.3 (from autotrain-advanced)
  Downloading xgboost-2.1.3-py3-none-
manylinux_2_28_x86_64.whl.metadata (2.1 kB)
Collecting huggingface-hub==0.27.0 (from autotrain-advanced)
  Downloading huggingface_hub-0.27.0-py3-none-any.whl.metadata (13 kB)
Collecting requests==2.32.3 (from autotrain-advanced)
  Downloading requests-2.32.3-py3-none-any.whl.metadata (4.6 kB)
Collecting einops==0.8.0 (from autotrain-advanced)
  Downloading einops-0.8.0-py3-none-any.whl.metadata (12 kB)
Collecting packaging==24.2 (from autotrain-advanced)
  Downloading packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
Collecting cryptography==44.0.0 (from autotrain-advanced)
  Downloading cryptography-44.0.0-cp39-abi3-
manylinux_2_28_x86_64.whl.metadata (5.7 kB)
Collecting nvitop==1.3.2 (from autotrain-advanced)
  Downloading nvitop-1.3.2-py3-none-any.whl.metadata (78 kB)


---


78.2/78.2 kB 5.9 MB/s eta
0:00:00
autotrain-advanced)
  Downloading tensorboard-2.18.0-py3-none-any.whl.metadata (1.6 kB)
Collecting peft==0.14.0 (from autotrain-advanced)
  Downloading peft-0.14.0-py3-none-any.whl.metadata (13 kB)
Collecting trl==0.13.0 (from autotrain-advanced)
  Downloading trl-0.13.0-py3-none-any.whl.metadata (11 kB)
Collecting tiktoken==0.8.0 (from autotrain-advanced)
  Downloading tiktoken-0.8.0-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.6 kB)
Collecting transformers==4.48.0 (from autotrain-advanced)
  Downloading transformers-4.48.0-py3-none-any.whl.metadata (44 kB)


---


44.4/44.4 kB 3.6 MB/s eta
0:00:00
autotrain-advanced)

```

```

    Downloading accelerate-1.2.1-py3-none-any.whl.metadata (19 kB)
Collecting rouge-score==0.1.2 (from autotrain-advanced)
    Downloading rouge_score-0.1.2.tar.gz (17 kB)
    Preparing metadata (setup.py) ... autotrain-advanced)
    Downloading py7zr-0.22.0-py3-none-any.whl.metadata (16 kB)
Collecting fastapi==0.115.6 (from autotrain-advanced)
    Downloading fastapi-0.115.6-py3-none-any.whl.metadata (27 kB)
Collecting uvicorn==0.34.0 (from autotrain-advanced)
    Downloading uvicorn-0.34.0-py3-none-any.whl.metadata (6.5 kB)
Collecting python-multipart==0.0.20 (from autotrain-advanced)
    Downloading python_multipart-0.0.20-py3-none-any.whl.metadata (1.8
kB)
Collecting pydantic==2.10.4 (from autotrain-advanced)
    Downloading pydantic-2.10.4-py3-none-any.whl.metadata (29 kB)
Collecting hf-transfer (from autotrain-advanced)
    Downloading hf_transfer-0.1.9-cp38-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (1.7 kB)
Collecting pyngrok==7.2.1 (from autotrain-advanced)
    Downloading pyngrok-7.2.1-py3-none-any.whl.metadata (8.3 kB)
Collecting authlib==1.4.0 (from autotrain-advanced)
    Downloading Authlib-1.4.0-py2.py3-none-any.whl.metadata (3.9 kB)
Requirement already satisfied: itsdangerous==2.2.0 in
/usr/local/lib/python3.12/dist-packages (from autotrain-advanced)
(2.2.0)
Collecting sequeval==1.2.2 (from autotrain-advanced)
    Downloading sequeval-1.2.2.tar.gz (43 kB)
----- 43.6/43.6 kB 3.5 MB/s eta
0:00:00
etadata (setup.py) ... ent already satisfied: httpx==0.28.1 in
/usr/local/lib/python3.12/dist-packages (from autotrain-advanced)
(0.28.1)
Collecting pyyaml==6.0.2 (from autotrain-advanced)
    Downloading PyYAML-6.0.2-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (2.1 kB)
Collecting timm==1.0.12 (from autotrain-advanced)
    Downloading timm-1.0.12-py3-none-any.whl.metadata (51 kB)
----- 51.6/51.6 kB 3.9 MB/s eta
0:00:00
etrics==1.6.0 (from autotrain-advanced)
    Downloading torchmetrics-1.6.0-py3-none-any.whl.metadata (20 kB)
Collecting pycocotools==2.0.8 (from autotrain-advanced)
    Downloading pycocotools-2.0.8-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (1.1 kB)
Collecting sentence-transformers==3.3.1 (from autotrain-advanced)
    Downloading sentence_transformers-3.3.1-py3-none-any.whl.metadata
(10 kB)
Collecting bitsandbytes==0.45.0 (from autotrain-advanced)
    Downloading bitsandbytes-0.45.0-py3-none-
manylinux_2_24_x86_64.whl.metadata (2.9 kB)

```

Requirement already satisfied: numpy<3.0.0,>=1.17 in
/usr/local/lib/python3.12/dist-packages (from accelerate==1.2.1->autotrain-advanced) (2.0.2)

Requirement already satisfied: psutil in
/usr/local/lib/python3.12/dist-packages (from accelerate==1.2.1->autotrain-advanced) (5.9.5)

Requirement already satisfied: torch>=1.10.0 in
/usr/local/lib/python3.12/dist-packages (from accelerate==1.2.1->autotrain-advanced) (2.11.0+cpu)

Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from accelerate==1.2.1->autotrain-advanced) (0.8.0)

Requirement already satisfied: scipy>=1.10.0 in
/usr/local/lib/python3.12/dist-packages (from albumentations==1.4.23->autotrain-advanced) (1.16.3)

Collecting albucore==0.0.21 (from albumentations==1.4.23->autotrain-advanced)

Downloading albucore-0.0.21-py3-none-any.whl.metadata (5.3 kB)

Collecting eval-type-backport (from albumentations==1.4.23->autotrain-advanced)

Downloading eval_type_backport-0.4.0-py3-none-any.whl.metadata (3.1 kB)

Requirement already satisfied: opencv-python-headless>=4.9.0.80 in
/usr/local/lib/python3.12/dist-packages (from albumentations==1.4.23->autotrain-advanced) (4.13.0.92)

Requirement already satisfied: typing_extensions>=4.8.0 in
/usr/local/lib/python3.12/dist-packages (from bitsandbytes==0.45.0->autotrain-advanced) (4.15.0)

Requirement already satisfied: cffi>=1.12 in
/usr/local/lib/python3.12/dist-packages (from cryptography==44.0.0->autotrain-advanced) (2.0.0)

Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (from evaluate==0.4.3->autotrain-advanced) (0.3.8)

Requirement already satisfied: xxhash in
/usr/local/lib/python3.12/dist-packages (from evaluate==0.4.3->autotrain-advanced) (3.7.1)

Requirement already satisfied: multiprocessing in
/usr/local/lib/python3.12/dist-packages (from evaluate==0.4.3->autotrain-advanced) (0.70.16)

Requirement already satisfied: fsspec>=2021.05.0 in
/usr/local/lib/python3.12/dist-packages (from fsspec[http]>=2021.05.0->evaluate==0.4.3->autotrain-advanced) (2025.3.0)

Collecting starlette<0.42.0,>=0.40.0 (from fastapi==0.115.6->autotrain-advanced)

Downloading starlette-0.41.3-py3-none-any.whl.metadata (6.0 kB)

Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx==0.28.1->autotrain-advanced) (4.14.0)

Requirement already satisfied: certifi in

```
/usr/local/lib/python3.12/dist-packages (from httpx==0.28.1-
>autotrain-advanced) (2026.6.17)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx==0.28.1-
>autotrain-advanced) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx==0.28.1->autotrain-advanced) (3.18)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.27.0-
>autotrain-advanced) (3.29.4)
Requirement already satisfied: click<9.0.0,>=8.1.3 in
/usr/local/lib/python3.12/dist-packages (from jiwer==3.0.5->autotrain-
advanced) (8.4.2)
Collecting rapidfuzz<4,>=3 (from jiwer==3.0.5->autotrain-advanced)
  Downloading rapidfuzz-3.14.5-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (12 kB)
Requirement already satisfied: regex>=2021.8.3 in
/usr/local/lib/python3.12/dist-packages (from nltk==3.9.1->autotrain-
advanced) (2025.11.3)
Collecting nvidia-ml-py<12.536.0a0,>=11.450.51 (from nvitop==1.3.2-
>autotrain-advanced)
  Downloading nvidia_ml_py-12.535.161-py3-none-any.whl.metadata (8.6
kB)
Requirement already satisfied: cachetools>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from nvitop==1.3.2-
>autotrain-advanced) (6.2.6)
Requirement already satisfied: termcolor>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from nvitop==1.3.2-
>autotrain-advanced) (3.3.0)
Collecting alembic>=1.5.0 (from optuna==4.1.0->autotrain-advanced)
  Downloading alembic-1.18.5-py3-none-any.whl.metadata (7.2 kB)
Collecting colorlog (from optuna==4.1.0->autotrain-advanced)
  Downloading colorlog-6.10.1-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: sqlalchemy>=1.4.2 in
/usr/local/lib/python3.12/dist-packages (from optuna==4.1.0-
>autotrain-advanced) (2.0.51)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas==2.2.3-
>autotrain-advanced) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas==2.2.3-
>autotrain-advanced) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas==2.2.3-
>autotrain-advanced) (2026.2)
Collecting texttable (from py7zr==0.22.0->autotrain-advanced)
  Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)
Requirement already satisfied: pycryptodomex>=3.16.0 in
/usr/local/lib/python3.12/dist-packages (from py7zr==0.22.0-
```

```
>autotrain-advanced) (3.23.0)
Collecting pyzstd>=0.15.9 (from py7zr==0.22.0->autotrain-advanced)
  Downloading pyzstd-0.19.1-py3-none-any.whl.metadata (2.6 kB)
Collecting pyppmd<1.2.0,>=1.1.0 (from py7zr==0.22.0->autotrain-advanced)
  Downloading pyppmd-1.1.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.5 kB)
Collecting pybcj<1.1.0,>=1.0.0 (from py7zr==0.22.0->autotrain-advanced)
  Downloading pybcj-1.0.7-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (3.9 kB)
Collecting multivolumefile>=0.2.3 (from py7zr==0.22.0->autotrain-advanced)
  Downloading multivolumefile-0.2.3-py3-none-any.whl.metadata (6.3 kB)
Collecting inflate64<1.1.0,>=1.0.0 (from py7zr==0.22.0->autotrain-advanced)
  Downloading inflate64-1.0.4-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (4.4 kB)
Requirement already satisfied: brotli>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from py7zr==0.22.0->autotrain-advanced) (1.2.0)
Requirement already satisfied: matplotlib>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from pycocotools==2.0.8->autotrain-advanced) (3.10.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic==2.10.4->autotrain-advanced) (0.7.0)
Collecting pydantic-core==2.27.2 (from pydantic==2.10.4->autotrain-advanced)
  Downloading pydantic_core-2.27.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.6 kB)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests==2.32.3->autotrain-advanced) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests==2.32.3->autotrain-advanced) (2.5.0)
Requirement already satisfied: absl-py in /usr/local/lib/python3.12/dist-packages (from rouge-score==0.1.2->autotrain-advanced) (1.4.0)
Requirement already satisfied: six>=1.14.0 in /usr/local/lib/python3.12/dist-packages (from rouge-score==0.1.2->autotrain-advanced) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn==1.6.0->autotrain-advanced) (3.6.0)
Requirement already satisfied: grpcio>=1.48.2 in
```

```
/usr/local/lib/python3.12/dist-packages (from tensorboard==2.18.0-
>autotrain-advanced) (1.81.1)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.12/dist-packages (from tensorboard==2.18.0-
>autotrain-advanced) (3.10.2)
Requirement already satisfied: protobuf!=4.24.0,>=3.19.6 in
/usr/local/lib/python3.12/dist-packages (from tensorboard==2.18.0-
>autotrain-advanced) (5.29.6)
Requirement already satisfied: setuptools>=41.0.0 in
/usr/local/lib/python3.12/dist-packages (from tensorboard==2.18.0-
>autotrain-advanced) (75.2.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0
in /usr/local/lib/python3.12/dist-packages (from tensorboard==2.18.0-
>autotrain-advanced) (0.7.2)
Requirement already satisfied: torchvision in
/usr/local/lib/python3.12/dist-packages (from timm==1.0.12->autotrain-
advanced) (0.26.0+cpu)
Collecting lightning-utilities>=0.8.0 (from torchmetrics==1.6.0-
>autotrain-advanced)
  Downloading lightning_utilities-0.15.3-py3-none-any.whl.metadata
(5.5 kB)
Collecting tokenizers<0.22,>=0.21 (from transformers==4.48.0-
>autotrain-advanced)
  Downloading tokenizers-0.21.4-cp39-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.7 kB)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-
packages (from trl==0.13.0->autotrain-advanced) (13.9.4)
Requirement already satisfied: h11>=0.8 in
/usr/local/lib/python3.12/dist-packages (from uvicorn==0.34.0-
>autotrain-advanced) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.1.1 in
/usr/local/lib/python3.12/dist-packages (from werkzeug==3.1.3-
>autotrain-advanced) (3.0.3)
Requirement already satisfied: nvidia-nccl-cu12 in
/usr/local/lib/python3.12/dist-packages (from xgboost==2.1.3-
>autotrain-advanced) (2.30.7)
Requirement already satisfied: stringzilla>=3.10.4 in
/usr/local/lib/python3.12/dist-packages (from albucore==0.0.21-
>albumentations==1.4.23->autotrain-advanced) (4.6.2)
Requirement already satisfied: simsimd>=5.9.2 in
/usr/local/lib/python3.12/dist-packages (from albucore==0.0.21-
>albumentations==1.4.23->autotrain-advanced) (6.5.16)
Requirement already satisfied: pyarrow>=15.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets~=3.2.0-
>datasets[vision]~=3.2.0->autotrain-advanced) (18.1.0)
Collecting fsspec>=2021.05.0 (from fsspec[http]>=2021.05.0-
>evaluate==0.4.3->autotrain-advanced)
  Downloading fsspec-2024.9.0-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: aiohttp in
```

```
/usr/local/lib/python3.12/dist-packages (from datasets~=3.2.0-
>datasets[vision]~=3.2.0->autotrain-advanced) (3.14.1)
Requirement already satisfied: Mako in /usr/lib/python3/dist-packages
(from alembic>=1.5.0->optuna==4.1.0->autotrain-advanced) (1.1.3)
Requirement already satisfied: pycparser in
/usr/local/lib/python3.12/dist-packages (from cffi>=1.12-
>cryptography==44.0.0->autotrain-advanced) (3.0)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced) (2.6.2)
Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced)
(26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced) (0.5.2)
Requirement already satisfied: yarll<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp-
>datasets~=3.2.0->datasets[vision]~=3.2.0->autotrain-advanced)
(1.24.2)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=2.1.0-
>pycocotools==2.0.8->autotrain-advanced) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=2.1.0-
>pycocotools==2.0.8->autotrain-advanced) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=2.1.0-
>pycocotools==2.0.8->autotrain-advanced) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=2.1.0-
>pycocotools==2.0.8->autotrain-advanced) (1.5.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=2.1.0-
>pycocotools==2.0.8->autotrain-advanced) (3.3.2)
Collecting backports-zstd>=1.0.0 (from pyzstd>=0.15.9->py7zr==0.22.0-
>autotrain-advanced)
  Downloading backports_zstd-1.6.0-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.m
```



```

etadata (6.8 kB)
Requirement already satisfied: greenlet>=1 in
/usr/local/lib/python3.12/dist-packages (from sqlalchemy>=1.4.2-
>optuna==4.1.0->autotrain-advanced) (3.5.2)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.10.0-
>accelerate==1.2.1->autotrain-advanced) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.10.0-
>accelerate==1.2.1->autotrain-advanced) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.10.0-
>accelerate==1.2.1->autotrain-advanced) (3.1.6)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich->trl==0.13.0-
>autotrain-advanced) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich->trl==0.13.0-
>autotrain-advanced) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich->trl==0.13.0->autotrain-advanced) (0.1.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=1.10.0->accelerate==1.2.1->autotrain-advanced) (1.3.0)
Downloading autotrain_advanced-0.8.36-py3-none-any.whl (341 kB)
_____ 341.3/341.3 kB 23.0 MB/s eta
0:00:00
_____ 336.4/336.4 kB 20.0 MB/s eta
0:00:00
entations-1.4.23-py3-none-any.whl (269 kB)
_____ 269.9/269.9 kB 16.3 MB/s eta
0:00:00
_____ 225.5/225.5 kB 15.4 MB/s eta
0:00:00
anylinux_2_24_x86_64.whl (69.1 MB)
_____ 69.1/69.1 MB 10.7 MB/s eta
0:00:00
anylinux_2_28_x86_64.whl (4.2 MB)
_____ 4.2/4.2 MB 97.9 MB/s eta
0:00:00
_____ 43.2/43.2 kB 2.5 MB/s eta
0:00:00
_____ 84.0/84.0 kB 6.1 MB/s eta
0:00:00
_____ 94.8/94.8 kB 7.3 MB/s eta
0:00:00
_____ 450.5/450.5 kB 31.0 MB/s eta
0:00:00

```

0:00:00	301.8/301.8 kB 19.4 MB/s eta
0:00:00	61.6/61.6 kB 5.0 MB/s eta
0:00:00	215.4/215.4 kB 15.1 MB/s eta
0:00:00	364.4/364.4 kB 25.4 MB/s eta
0:00:00	65.5/65.5 kB 4.2 MB/s eta
0:00:00	anylinux_2_17_x86_64.manylinux2014_x86_64.whl (12.7 MB)
0:00:00	12.7/12.7 MB 91.3 MB/s eta
0:00:00	374.8/374.8 kB 27.4 MB/s eta
0:00:00	anylinux_2_28_x86_64.whl (4.4 MB)
0:00:00	4.4/4.4 MB 93.4 MB/s eta
0:00:00	67.9/67.9 kB 5.4 MB/s eta
0:00:00	anylinux_2_17_x86_64.manylinux2014_x86_64.whl (443 kB)
0:00:00	443.5/443.5 kB 30.7 MB/s eta
0:00:00	431.8/431.8 kB 27.1 MB/s eta
0:00:00	ultipart-0.0.20-py3-none-any.whl (24 kB)
0:00:00	Downloading PyYAML-6.0.2-cp312-cp312-
0:00:00	manylinux_2_17_x86_64.manylinux2014_x86_64.whl (767 kB)
0:00:00	767.5/767.5 kB 44.6 MB/s eta
0:00:00	64.9/64.9 kB 5.4 MB/s eta
0:00:00	oses-0.1.1-py3-none-any.whl (897 kB)
0:00:00	897.5/897.5 kB 46.5 MB/s eta
0:00:00	anylinux_2_17_x86_64.manylinux2014_x86_64.whl (13.1 MB)
0:00:00	13.1/13.1 MB 79.8 MB/s eta
0:00:00	ers-3.3.1-py3-none-any.whl (268 kB)
0:00:00	268.8/268.8 kB 19.1 MB/s eta
0:00:00	anylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.3 MB)
0:00:00	1.3/1.3 MB 56.9 MB/s eta
0:00:00	5.5/5.5 MB 113.4 MB/s eta
0:00:00	anylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.2 MB)
0:00:00	1.2/1.2 MB 56.1 MB/s eta

```

m-1.0.12-py3-none-any.whl (2.4 MB)
----- 2.4/2.4 MB 65.7 MB/s eta
0:00:00
etrics-1.6.0-py3-none-any.whl (926 kB)
----- 926.4/926.4 kB 46.3 MB/s eta
0:00:00
-4.67.1-py3-none-any.whl (78 kB)
----- 78.5/78.5 kB 5.3 MB/s eta
0:00:00
ers-4.48.0-py3-none-any.whl (9.7 MB)
----- 9.7/9.7 MB 93.3 MB/s eta
0:00:00
----- 293.4/293.4 kB 19.9 MB/s eta
0:00:00
----- 62.3/62.3 kB 4.1 MB/s eta
0:00:00
----- 224.5/224.5 kB 16.5 MB/s eta
0:00:00
anylinux_2_28_x86_64.whl (153.9 MB)
----- 153.9/153.9 MB 7.4 MB/s eta
0:00:00
anylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0 MB)
----- 2.0/2.0 MB 40.3 MB/s eta
0:00:00
anylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.0 MB)
----- 5.0/5.0 MB 51.1 MB/s eta
0:00:00
----- 480.6/480.6 kB 31.7 MB/s eta
0:00:00
anylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.6 MB)
----- 3.6/3.6 MB 46.5 MB/s eta
0:00:00
bic-1.18.5-py3-none-any.whl (264 kB)
----- 264.7/264.7 kB 17.2 MB/s eta
0:00:00
----- 179.3/179.3 kB 16.1 MB/s eta
0:00:00
anylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl
(100 kB)
----- 100.6/100.6 kB 9.8 MB/s eta
0:00:00
ultivolume-0.2.3-py3-none-any.whl (17 kB)
Downloading nvidia_ml_py-12.535.161-py3-none-any.whl (37 kB)
Downloading pybcj-1.0.7-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl
(51 kB)
----- 51.5/51.5 kB 4.8 MB/s eta
0:00:00
d-1.1.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(142 kB)

```

```

142.8/142.8 kB 12.6 MB/s eta
0:00:00
anylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (3.1 MB)
3.1/3.1 MB 46.5 MB/s eta
0:00:00
73.2/73.2 kB 6.9 MB/s eta
0:00:00
anylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.1 MB)
3.1/3.1 MB 52.0 MB/s eta
0:00:00
anylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl
(495 kB)
495.6/495.6 kB 24.8 MB/s eta
0:00:00
e=ipadic-1.0.0-py3-none-any.whl size=13556704
sha256=7ec51de0761c8eec66f0e380125dada6430746088537585d6dfd1a5f56d3354
0
Stored in directory:
/root/.cache/pip/wheels/93/8b/55/dd5978a069678c372520847cf84ba2ec539cb
41917c00a2206
Building wheel for rouge-score (setup.py) ... e=rouge_score-0.1.2-
py3-none-any.whl size=24934
sha256=5f49aba2f35c98ebbe5e5a8da36bb769853c8b1d93e51d92373f25aa46ecbce
a
Stored in directory:
/root/.cache/pip/wheels/85/9d/af/01feefbe7d55ef5468796f0c68225b6788e85
d9d0a281e7a70
Building wheel for sequeval (setup.py) ... e=sequeval-1.2.2-py3-none-
any.whl size=16162
sha256=288e28122e8d180841762520a974f20d8d67e13ee0e79bb37ae59b325e0ef42
c
Stored in directory:
/root/.cache/pip/wheels/5f/b8/73/0b2c1a76b701a677653dd79ece07cfabd7457
989dbfbdc8d7
Successfully built ipadic rouge-score sequeval
Installing collected packages: texttable, sentencepiece, pysqlite3-
binary, nvidia-ml-py, ipadic, werkzeug, uvicorn, tqdm, requests,
rapidfuzz, pyyaml, python-multipart, pyppmd, pydantic-core, pybcj,
Pillow, packaging, nvitop, multivolumefile, loguru, joblib, inflate64,
hf-transfer, fsspec, eval-type-backport, einops, colorlog, backports-
zstd, xgboost, tiktoken, tensorboard, starlette, scikit-learn,
sacremoses, pyzstd, pyngrok, pydantic, pandas, lightning-utilities,
jiwer, huggingface-hub, cryptography, alembic, albucore, torchmetrics,
tokenizers, sequeval, rouge-score, pycocotools, py7zr, optuna, fastapi,
bitsandbytes, authlib, alumentations, accelerate, transformers, timm,
datasets, trl, sentence-transformers, peft, evaluate, autotrain-
advanced
Attempting uninstall: sentencepiece
Found existing installation: sentencepiece 0.2.1

```

```
Uninstalling sentencepiece-0.2.1:
  Successfully uninstalled sentencepiece-0.2.1
Attempting uninstall: werkzeug
  Found existing installation: Werkzeug 3.1.8
  Uninstalling Werkzeug-3.1.8:
    Successfully uninstalled Werkzeug-3.1.8
Attempting uninstall: uvicorn
  Found existing installation: uvicorn 0.49.0
  Uninstalling uvicorn-0.49.0:
    Successfully uninstalled uvicorn-0.49.0
Attempting uninstall: tqdm
  Found existing installation: tqdm 4.67.3
  Uninstalling tqdm-4.67.3:
    Successfully uninstalled tqdm-4.67.3
Attempting uninstall: requests
  Found existing installation: requests 2.32.4
  Uninstalling requests-2.32.4:
    Successfully uninstalled requests-2.32.4
Attempting uninstall: pyyaml
  Found existing installation: PyYAML 6.0.3
  Uninstalling PyYAML-6.0.3:
    Successfully uninstalled PyYAML-6.0.3
Attempting uninstall: python-multipart
  Found existing installation: python-multipart 0.0.32
  Uninstalling python-multipart-0.0.32:
    Successfully uninstalled python-multipart-0.0.32
Attempting uninstall: pydantic-core
  Found existing installation: pydantic_core 2.46.4
  Uninstalling pydantic_core-2.46.4:
    Successfully uninstalled pydantic_core-2.46.4
Attempting uninstall: Pillow
  Found existing installation: pillow 11.3.0
  Uninstalling pillow-11.3.0:
    Successfully uninstalled pillow-11.3.0
Attempting uninstall: packaging
  Found existing installation: packaging 26.2
  Uninstalling packaging-26.2:
    Successfully uninstalled packaging-26.2
Attempting uninstall: joblib
  Found existing installation: joblib 1.5.3
  Uninstalling joblib-1.5.3:
    Successfully uninstalled joblib-1.5.3
Attempting uninstall: fsspec
  Found existing installation: fsspec 2025.3.0
  Uninstalling fsspec-2025.3.0:
    Successfully uninstalled fsspec-2025.3.0
Attempting uninstall: einops
  Found existing installation: einops 0.8.2
  Uninstalling einops-0.8.2:
```

```
    Successfully uninstalled einops-0.8.2
Attempting uninstall: xgboost
  Found existing installation: xgboost 3.3.0
  Uninstalling xgboost-3.3.0:
    Successfully uninstalled xgboost-3.3.0
Attempting uninstall: tiktoken
  Found existing installation: tiktoken 0.13.0
  Uninstalling tiktoken-0.13.0:
    Successfully uninstalled tiktoken-0.13.0
Attempting uninstall: tensorboard
  Found existing installation: tensorboard 2.20.0
  Uninstalling tensorboard-2.20.0:
    Successfully uninstalled tensorboard-2.20.0
Attempting uninstall: starlette
  Found existing installation: starlette 1.3.1
  Uninstalling starlette-1.3.1:
    Successfully uninstalled starlette-1.3.1
Attempting uninstall: scikit-learn
  Found existing installation: scikit-learn 1.6.1
  Uninstalling scikit-learn-1.6.1:
    Successfully uninstalled scikit-learn-1.6.1
Attempting uninstall: pydantic
  Found existing installation: pydantic 2.13.4
  Uninstalling pydantic-2.13.4:
    Successfully uninstalled pydantic-2.13.4
Attempting uninstall: pandas
  Found existing installation: pandas 2.2.2
  Uninstalling pandas-2.2.2:
    Successfully uninstalled pandas-2.2.2
Attempting uninstall: huggingface-hub
  Found existing installation: huggingface_hub 1.20.1
  Uninstalling huggingface_hub-1.20.1:
    Successfully uninstalled huggingface_hub-1.20.1
Attempting uninstall: cryptography
  Found existing installation: cryptography 49.0.0
  Uninstalling cryptography-49.0.0:
    Successfully uninstalled cryptography-49.0.0
Attempting uninstall: albucore
  Found existing installation: albucore 0.0.24
  Uninstalling albucore-0.0.24:
    Successfully uninstalled albucore-0.0.24
Attempting uninstall: tokenizers
  Found existing installation: tokenizers 0.22.2
  Uninstalling tokenizers-0.22.2:
    Successfully uninstalled tokenizers-0.22.2
Attempting uninstall: pycocotools
  Found existing installation: pycocotools 2.0.11
  Uninstalling pycocotools-2.0.11:
    Successfully uninstalled pycocotools-2.0.11
```

```
Attempting uninstall: fastapi
  Found existing installation: fastapi 0.138.0
  Uninstalling fastapi-0.138.0:
    Successfully uninstalled fastapi-0.138.0
Attempting uninstall: authlib
  Found existing installation: Authlib 1.7.2
  Uninstalling Authlib-1.7.2:
    Successfully uninstalled Authlib-1.7.2
Attempting uninstall: alumentations
  Found existing installation: alumentations 2.0.8
  Uninstalling alumentations-2.0.8:
    Successfully uninstalled alumentations-2.0.8
Attempting uninstall: accelerate
  Found existing installation: accelerate 1.14.0
  Uninstalling accelerate-1.14.0:
    Successfully uninstalled accelerate-1.14.0
Attempting uninstall: transformers
  Found existing installation: transformers 5.12.1
  Uninstalling transformers-5.12.1:
    Successfully uninstalled transformers-5.12.1
Attempting uninstall: timm
  Found existing installation: timm 1.0.27
  Uninstalling timm-1.0.27:
    Successfully uninstalled timm-1.0.27
Attempting uninstall: datasets
  Found existing installation: datasets 4.0.0
  Uninstalling datasets-4.0.0:
    Successfully uninstalled datasets-4.0.0
Attempting uninstall: sentence-transformers
  Found existing installation: sentence-transformers 5.6.0
  Uninstalling sentence-transformers-5.6.0:
    Successfully uninstalled sentence-transformers-5.6.0
Attempting uninstall: peft
  Found existing installation: peft 0.19.1
  Uninstalling peft-0.19.1:
    Successfully uninstalled peft-0.19.1
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3
which is incompatible.
google-colab 1.0.0 requires requests==2.32.4, but you have requests
2.32.3 which is incompatible.
google-genai 2.10.0 requires pydantic<3.0.0,>=2.12.5, but you have
pydantic 2.10.4 which is incompatible.
python-fasthtml 0.14.3 requires starlette>=1.0.1, but you have
starlette 0.41.3 which is incompatible.
google-adk 2.3.0 requires authlib<2,>=1.6.6, but you have authlib
1.4.0 which is incompatible.
```

google-adk 2.3.0 requires fastapi<1,>=0.133, but you have fastapi 0.115.6 which is incompatible.
 google-adk 2.3.0 requires pydantic<3,>=2.12, but you have pydantic 2.10.4 which is incompatible.
 google-adk 2.3.0 requires requests<3,>=2.32.4, but you have requests 2.32.3 which is incompatible.
 google-adk 2.3.0 requires starlette<2,>=1.0.1, but you have starlette 0.41.3 which is incompatible.
 joserfc 1.7.1 requires cryptography>=45.0.1, but you have cryptography 44.0.0 which is incompatible.
 gradio 6.19.0 requires huggingface-hub<2.0,>=1.2.0, but you have huggingface-hub 0.27.0 which is incompatible.
 gradio 6.19.0 requires starlette<2.0,>=1.0.1, but you have starlette 0.41.3 which is incompatible.
 diffusers 0.38.0 requires huggingface-hub<2.0,>=0.34.0, but you have huggingface-hub 0.27.0 which is incompatible.
 pyopenssl 26.3.0 requires cryptography<50,>=49.0.0, but you have cryptography 44.0.0 which is incompatible.
 gcsfs 2025.3.0 requires fsspec==2025.3.0, but you have fsspec 2024.9.0 which is incompatible.
 tensorflow 2.20.0 requires tensorboard~=2.20.0, but you have tensorboard 2.18.0 which is incompatible.
 Successfully installed Pillow-11.0.0 accelerate-1.2.1 albucore-0.0.21 alumentations-1.4.23 alembic-1.18.5 authlib-1.4.0 autotrain-advanced-0.8.36 backports-zstd-1.6.0 bitsandbytes-0.45.0 colorlog-6.10.1 cryptography-44.0.0 datasets-3.2.0 einops-0.8.0 eval-type-backport-0.4.0 evaluate-0.4.3 fastapi-0.115.6 fsspec-2024.9.0 hf-transfer-0.1.9 huggingface-hub-0.27.0 inflate64-1.0.4 ipadic-1.0.0 jiwer-3.0.5 joblib-1.4.2 lightning-utilities-0.15.3 loguru-0.7.3 multivolumefile-0.2.3 nvidia-ml-py-12.535.161 nvitop-1.3.2 optuna-4.1.0 packaging-24.2 pandas-2.2.3 peft-0.14.0 py7zr-0.22.0 pybcj-1.0.7 pycocotools-2.0.8 pydantic-2.10.4 pydantic-core-2.27.2 pyngrok-7.2.1 pyppmd-1.1.1 pysqlite3-binary-0.5.4.post2 python-multipart-0.0.20 pyyaml-6.0.2 pyzstd-0.19.1 rapidfuzz-3.14.5 requests-2.32.3 rouge-score-0.1.2 sacremoses-0.1.1 scikit-learn-1.6.0 sentence-transformers-3.3.1 sentencepiece-0.2.0 seqeval-1.2.2 starlette-0.41.3 tensorboard-2.18.0 texttable-1.7.0 tiktoken-0.8.0 timm-1.0.12 tokenizers-0.21.4 torchmetrics-1.6.0 tqdm-4.67.1 transformers-4.48.0 trl-0.13.0 uvicorn-0.34.0 werkzeug-3.1.3 xgboost-2.1.3

```
{"id":"717caddledf14033994ff0eb537379d5","pip_warning":{"packages":["packaging"]}}
```

□ Fix SQLite appliqué !

#Étape 3 : Formatage ChatML (Spécifique à SmolLM2)

```
from transformers import AutoTokenizer
import json
```



```

# 1. Charger le tokenizer
model_id = "HuggingFaceTB/SmolLM2-360M-Instruct"
tokenizer = AutoTokenizer.from_pretrained(model_id)

# 2. Transformer le JSONL vers le format 'text' attendu par AutoTrain
formatted_data = []
with open("train.jsonl", "r", encoding="utf-8") as f:
    for line in f:
        data = json.loads(line)
        # On applique le template de chat du modèle si nécessaire
        if "messages" in data:
            full_text =
tokenizer.apply_chat_template(data["messages"], tokenize=False)
        elif "text" in data:
            full_text = data["text"]
        else:
            print(f"Warning: Skipping line with unexpected format:
{line.strip()}")
            continue
        formatted_data.append({"text": full_text})

# 3. Écraser le fichier train.jsonl avec le nouveau format
with open("train.jsonl", "w", encoding="utf-8") as f:
    for entry in formatted_data:
        f.write(json.dumps(entry, ensure_ascii=False) + "\n")

print(f"✅ Correction terminée ! Fichier prêt avec
{len(formatted_data)} lignes au format ChatML.")

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
warnings.warn(

{"model_id": "a2ee2e4b06c94a8280022167dc323439", "version_major": 2, "vers
ion_minor": 0}

{"model_id": "1572f4e96c0142a8bc2cd8f22328d068", "version_major": 2, "vers
ion_minor": 0}

{"model_id": "d52241913a2c4463baa951bcd0cc4bef", "version_major": 2, "vers
ion_minor": 0}

{"model_id": "f702b990778547aaa1e960ec9d676ddc", "version_major": 2, "vers
ion_minor": 0}

```

```
{"model_id":"f7b86329f1bc45e98ac045bb7147d834","version_major":2,"version_minor":0}
```

□ Correction terminée ! Fichier prêt avec 51357 lignes au format ChatML.

#Étape 4 : Lancer le Fine-Tuning

On supprime les anciens dossiers pour éviter les erreurs

```
!rm -rf professional-assistant
```

```
!autotrain llm --train \  
  --model "HuggingFaceTB/SmolLM2-360M-Instruct" \  
  --data-path . \  
  --train-split "train" \  
  --text-column "text" \  
  --lr 1e-4 \  
  --batch-size 4 \  
  --epochs 1 \  
  --trainer sft \  
  --model-max-length 512 \  
  --project-name "professional-assistant" \  
  --mixed-precision "fp16" \  
  --block-size 512 \  
  --disable-gradient-checkpointing
```

```
INFO      | 2026-07-01 17:14:08 | autotrain.cli.run_llm:run:136 -  
Running LLM
```

```
WARNING   | 2026-07-01 17:14:08 |  
autotrain.trainers.common:__init__:286 - Parameters supplied but not  
used: config, backend, inference, func, deploy, version, train  
Saving the dataset (1/1 shards): 100% 51357/51357 [00:00<00:00,  
1142766.27 examples/s]
```

```
Saving the dataset (1/1 shards): 100% 51357/51357 [00:00<00:00,  
1299079.52 examples/s]
```

```
INFO      | 2026-07-01 17:14:10 | autotrain.backends.local:create:20 -  
Starting local training...
```

```
WARNING   | 2026-07-01 17:14:10 |  
autotrain.commands.get_accelerate_command:58 - No GPU found. Forcing  
training on CPU. This will be super slow!
```

```
INFO      | 2026-07-01 17:14:10 | autotrain.commands.launch_command:514  
- ['accelerate', 'launch', '--cpu', '-m', 'autotrain.trainers.clm',  
  '--training_config', 'professional-assistant/training_params.json']
```

```
INFO      | 2026-07-01 17:14:10 | autotrain.commands.launch_command:515  
- {'model': 'HuggingFaceTB/SmolLM2-360M-Instruct', 'project_name':  
'professional-assistant', 'data_path':  
'professional-assistant/autotrain-data', 'train_split': 'train',  
'valid_split': None, 'add_eos_token': True, 'block_size': 512,  
'model_max_length': 512, 'padding': 'right', 'trainer': 'sft',  
'use_flash_attention_2': False, 'log': 'none',
```

```
'disable_gradient_checkpointing': True, 'logging_steps': -1,
'eval_strategy': 'epoch', 'save_total_limit': 1,
'auto_find_batch_size': False, 'mixed_precision': 'fp16', 'lr':
0.0001, 'epochs': 1, 'batch_size': 4, 'warmup_ratio': 0.1,
'gradient_accumulation': 4, 'optimizer': 'adamw_torch', 'scheduler':
'linear', 'weight_decay': 0.0, 'max_grad_norm': 1.0, 'seed': 42,
'chat_template': None, 'quantization': 'int4', 'target_modules': 'all-
linear', 'merge_adapter': False, 'peft': False, 'lora_r': 16,
'lora_alpha': 32, 'lora_dropout': 0.05, 'model_ref': None, 'dpo_beta':
0.1, 'max_prompt_length': 128, 'max_completion_length': None,
'prompt_text_column': 'autotrain_prompt', 'text_column':
'autotrain_text', 'rejected_text_column': 'autotrain_rejected_text',
'push_to_hub': False, 'username': None, 'token': None, 'unsloth':
False, 'distributed_backend': None}
```

The following values were not passed to `accelerate launch` and had defaults used instead:

```
`--num_processes` was set to a value of `0`
`--num_machines` was set to a value of `1`
`--mixed_precision` was set to a value of `no`
`--dynamo_backend` was set to a value of `no`
```

To avoid this warning pass in values for each of the problematic parameters or run `accelerate config`.

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.train_clm_sft:train:11 - Starting SFT
training...
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:process_input_data:487 - loading dataset
from disk
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:process_input_data:550 - Train data:
```

```
Dataset({
  features: ['autotrain_text'],
  num_rows: 51357
})
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:process_input_data:551 - Valid data: None
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:configure_logging_steps:671 - configuring
logging steps
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:configure_logging_steps:684 - Logging
steps: 25
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:configure_training_args:723 - configuring
training args
```

```
INFO      | 2026-07-01 17:14:37 |
```

```
autotrain.trainers.clm.utils:configure_block_size:801 - Using block
size 512
```

```
config.json: 100% 846/846 [00:00<00:00, 5.49MB/s]
```

```

INFO      | 2026-07-01 17:14:37 |
autotrain.trainers.clm.utils:get_model:877 - Can use unsloth: False
WARNING   | 2026-07-01 17:14:37 |
autotrain.trainers.clm.utils:get_model:919 - Unsloth not available,
continuing without it...
INFO      | 2026-07-01 17:14:37 |
autotrain.trainers.clm.utils:get_model:921 - loading model config...
INFO      | 2026-07-01 17:14:38 |
autotrain.trainers.clm.utils:get_model:929 - loading model...
model.safetensors: 100% 724M/724M [00:09<00:00, 73.6MB/s]
generation_config.json: 100% 132/132 [00:00<00:00, 994kB/s]
INFO      | 2026-07-01 17:14:50 |
autotrain.trainers.clm.utils:get_model:960 - model dtype:
torch.float32
INFO      | 2026-07-01 17:14:50 |
autotrain.trainers.clm.train_clm_sft:train:39 - creating trainer
Generating train split: 0 examples [00:00, ? examples/s]Token indices
sequence length is longer than the specified maximum sequence length
for this model (607 > 512). Running this sequence through the model
will result in indexing errors
Generating train split: 18566 examples [00:32, 565.94 examples/s]
INFO      | 2026-07-01 17:15:23 |
autotrain.trainers.common:on_train_begin:386 - Starting to train...
  0% 0/1160 [00:00<?,
?it/s]/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloa
der.py:775: UserWarning: 'pin_memory' argument is set as true but no
accelerator is found, then device pinned memory won't be used.
  super().__init__(loader)
Traceback (most recent call last):
  File "/usr/local/bin/accelerate", line 8, in <module>
    sys.exit(main())
            ^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/accelerate/commands/accelerat
e_cli.py", line 48, in main
    args.func(args)
  File
"/usr/local/lib/python3.12/dist-packages/accelerate/commands/launch.py",
line 1168, in launch_command
    simple_launcher(args)
  File
"/usr/local/lib/python3.12/dist-packages/accelerate/commands/launch.py",
line 763, in simple_launcher
    raise subprocess.CalledProcessError(returncode=process.returncode,
cmd=cmd)
subprocess.CalledProcessError: Command '['/usr/bin/python3', '-m',
'autotrain.trainers.clm', '--training_config', 'professional-
assistant/training_params.json']' died with <Signals.SIGKILL: 9>.

```

INFO | 2026-07-01 17:17:18 | autotrain.cli.run_llm:run:141 - Job ID: 4959

#l'étape finale pour convertir votre modèle entraîné au format GGUF.

Installation de llama.cpp

1. Cloner le dépôt officiel

!git clone https://github.com/ggerganov/llama.cpp

2. Installer les bibliothèques nécessaires

!pip install -r llama.cpp/requirements.txt

!pip install huggingface_hub

Cloning into 'llama.cpp'...

remote: Enumerating objects: 101547, done.ote: Counting objects: 100% (58/58), done.ote: Compressing objects: 100% (53/53), done.ote: Total 101547 (delta 23), reused 6 (delta 5), pack-reused 101489 (from 3)ple,

https://download.pytorch.org/whl/cpu,

https://download.pytorch.org/whl/nightly,

https://download.pytorch.org/whl/cpu,

https://download.pytorch.org/whl/nightly,

https://download.pytorch.org/whl/cpu,

https://download.pytorch.org/whl/nightly

Ignoring torch: markers 'platform_machine == "s390x"' don't match your environment

Ignoring torch: markers 'platform_machine == "s390x"' don't match your environment

Collecting numpy~=1.26.4 (from -r

llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line 1))

Downloading numpy-1.26.4-cp312-cp312-

manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)

61.0/61.0 kB 2.9 MB/s eta

0:00:00

ent already satisfied: sentencepiece<0.3.0,>=0.1.98 in

/usr/local/lib/python3.12/dist-packages (from -r

llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line 2)) (0.2.0)

Collecting transformers==4.57.6 (from -r

llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line 4))

Downloading transformers-4.57.6-py3-none-any.whl.metadata (43 kB)

44.0/44.0 kB 2.9 MB/s eta

0:00:00

-r llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line 6))

Downloading gguf-0.19.0-py3-none-any.whl.metadata (4.1 kB)

Collecting protobuf<5.0.0,>=4.21.0 (from -r

llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line

```

7))
  Downloading protobuf-4.25.9-cp37-abi3-
manylinux2014_x86_64.whl.metadata (541 bytes)
Requirement already satisfied: torch==2.11.0 in
/usr/local/lib/python3.12/dist-packages (from -r
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(2.11.0+cpu)
Collecting aiohttp~=3.9.3 (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 1))
  Downloading https://download.pytorch.org/whl/nightly/aiohttp-3.9.5-
cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.3 MB)
----- 1.3/1.3 MB 52.1 MB/s eta
0:00:00
  -r llama.cpp/./requirements/requirements-tool_bench.txt (line 2))
  Downloading pytest-8.3.5-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: matplotlib~=3.10.0 in
/usr/local/lib/python3.12/dist-packages (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3))
(3.10.0)
Collecting openai~=2.14.0 (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5))
  Downloading openai-2.14.0-py3-none-any.whl.metadata (29 kB)
Requirement already satisfied: pandas~=2.2.3 in
/usr/local/lib/python3.12/dist-packages (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 6)) (2.2.3)
Collecting prometheus-client~=0.20.0 (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 7))
  Downloading prometheus_client-0.20.0-py3-none-any.whl.metadata (1.8
kB)
Requirement already satisfied: requests~=2.32.3 in
/usr/local/lib/python3.12/dist-packages (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 8))
(2.32.3)
Collecting wget~=3.2 (from -r llama.cpp/./requirements/requirements-
tool_bench.txt (line 9))
  Downloading wget-3.2.zip (10 kB)
  Preparing metadata (setup.py) ... -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 10))
  Downloading typer-0.24.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: seaborn~=0.13.2 in
/usr/local/lib/python3.12/dist-packages (from -r
llama.cpp/./requirements/requirements-tool_bench.txt (line 11))
(0.13.2)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (3.29.4)
Collecting huggingface-hub<1.0,>=0.34.0 (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line

```

```
4))
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (24.2)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (2025.11.3)
Collecting tokenizers<=0.23.0,>=0.22.0 (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4))
  Downloading tokenizers-0.22.2-cp39-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (0.8.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (4.67.1)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch==2.11.0->-r
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(4.15.0)
Requirement already satisfied: setuptools<82 in
/usr/local/lib/python3.12/dist-packages (from torch==2.11.0->-r
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch==2.11.0->-r
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch==2.11.0->-r
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch==2.11.0->-r
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch==2.11.0->-r
```

```
llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line 5))
(2024.9.0)
Requirement already satisfied: aiosignal>=1.1.2 in
/usr/local/lib/python3.12/dist-packages (from aiohttp~=3.9.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 1)) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp~=3.9.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 1))
(26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp~=3.9.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 1)) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp~=3.9.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 1)) (6.7.1)
Requirement already satisfied: yarl<2.0,>=1.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp~=3.9.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 1))
(1.24.2)
Requirement already satisfied: iniconfig in
/usr/local/lib/python3.12/dist-packages (from pytest~=8.3.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 2)) (2.3.0)
Requirement already satisfied: pluggy<2,>=1.5 in
/usr/local/lib/python3.12/dist-packages (from pytest~=8.3.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 2)) (1.6.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3)) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3))
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3))
(4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3)) (1.5.0)
Requirement already satisfied: pillow>=8 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3))
(11.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3)) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib~=3.10.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 3))
```



```
(2.9.0.post0)
Requirement already satisfied: anyio<5,>=3.5.0 in
/usr/local/lib/python3.12/dist-packages (from openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5))
(4.14.0)
Requirement already satisfied: distro<2,>=1.7.0 in
/usr/local/lib/python3.12/dist-packages (from openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5)) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5))
(0.28.1)
Requirement already satisfied: jiter<1,>=0.10.0 in
/usr/local/lib/python3.12/dist-packages (from openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5))
(0.15.0)
Requirement already satisfied: pydantic<3,>=1.9.0 in
/usr/local/lib/python3.12/dist-packages (from openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5))
(2.10.4)
Requirement already satisfied: sniffio in
/usr/local/lib/python3.12/dist-packages (from openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5)) (1.3.1)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas~=2.2.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 6))
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas~=2.2.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 6))
(2026.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests~=2.32.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 8)) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests~=2.32.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 8)) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests~=2.32.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 8)) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests~=2.32.3->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 8))
(2026.6.17)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer~=0.24.1->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 10))
(8.4.2)
Requirement already satisfied: shellingham>=1.3.0 in
```

```

/usr/local/lib/python3.12/dist-packages (from typer~=0.24.1->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 10))
(1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer~=0.24.1->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 10))
(13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer~=0.24.1->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 10))
(0.0.4)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>openai~=2.14.0->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 5)) (1.0.9)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->openai~=2.14.0->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 5))
(0.16.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers==4.57.6->-r
llama.cpp/./requirements/requirements-convert_legacy_llama.txt (line
4)) (1.5.1)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0-
>openai~=2.14.0->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 5)) (0.7.0)
Requirement already satisfied: pydantic-core==2.27.2 in
/usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0-
>openai~=2.14.0->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 5)) (2.27.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7-
>matplotlib~=3.10.0->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 3)) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer~=0.24.1->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 10)) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer~=0.24.1->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 10)) (2.20.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch==2.11.0->-r llama.cpp/./requirements/requirements-
convert_hf_to_gguf.txt (line 5)) (1.3.0)

```

```

Requirement already satisfied: propcache>=0.2.1 in
/usr/local/lib/python3.12/dist-packages (from yarl<2.0,>=1.0-
>aiohttp~=3.9.3->-r llama.cpp/./requirements/requirements-
tool_bench.txt (line 1)) (0.5.2)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch==2.11.0->-
r llama.cpp/./requirements/requirements-convert_hf_to_gguf.txt (line
5)) (3.0.3)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer~=0.24.1->-r
llama.cpp/./requirements/requirements-tool_bench.txt (line 10))
(0.1.2)
Downloading transformers-4.57.6-py3-none-any.whl (12.0 MB)
----- 12.0/12.0 MB 91.4 MB/s eta
0:00:00
py-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(18.0 MB)
----- 18.0/18.0 MB 72.0 MB/s eta
0:00:00
----- 118.5/118.5 kB 10.2 MB/s eta
0:00:00
anylinux2014_x86_64.whl (295 kB)
----- 295.2/295.2 kB 19.4 MB/s eta
0:00:00
----- 343.6/343.6 kB 23.8 MB/s eta
0:00:00
----- 1.1/1.1 MB 54.0 MB/s eta
0:00:00
ethesus_client-0.20.0-py3-none-any.whl (54 kB)
----- 54.5/54.5 kB 4.3 MB/s eta
0:00:00
----- 56.0/56.0 kB 4.9 MB/s eta
0:00:00
----- 566.4/566.4 kB 39.0 MB/s eta
0:00:00
anylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.3 MB)
----- 3.3/3.3 MB 78.0 MB/s eta
0:00:00
e=wget-3.2-py3-none-any.whl size=9655
sha256=df2be2b3457213a4412b890735c3e84a08e1421cd002c63c9824ce27a0fd00c
b
  Stored in directory:
/root/.cache/pip/wheels/01/46/3b/e29ffbe4ebe614ff224bad40fc6a5773a67a1
63251585a13a9
Successfully built wget
Installing collected packages: wget, pytest, protobuf, prometheus-
client, numpy, huggingface-hub, gguf, aiohttp, typer, tokenizers,
openai, transformers

```

```
Attempting uninstall: pytest
  Found existing installation: pytest 8.4.2
  Uninstalling pytest-8.4.2:
    Successfully uninstalled pytest-8.4.2
Attempting uninstall: protobuf
  Found existing installation: protobuf 5.29.6
  Uninstalling protobuf-5.29.6:
    Successfully uninstalled protobuf-5.29.6
Attempting uninstall: prometheus-client
  Found existing installation: prometheus_client 0.25.0
  Uninstalling prometheus_client-0.25.0:
    Successfully uninstalled prometheus_client-0.25.0
Attempting uninstall: numpy
  Found existing installation: numpy 2.0.2
  Uninstalling numpy-2.0.2:
    Successfully uninstalled numpy-2.0.2
Attempting uninstall: huggingface-hub
  Found existing installation: huggingface-hub 0.27.0
  Uninstalling huggingface-hub-0.27.0:
    Successfully uninstalled huggingface-hub-0.27.0
Attempting uninstall: aiohttp
  Found existing installation: aiohttp 3.14.1
  Uninstalling aiohttp-3.14.1:
    Successfully uninstalled aiohttp-3.14.1
Attempting uninstall: typer
  Found existing installation: typer 0.25.1
  Uninstalling typer-0.25.1:
    Successfully uninstalled typer-0.25.1
Attempting uninstall: tokenizers
  Found existing installation: tokenizers 0.21.4
  Uninstalling tokenizers-0.21.4:
    Successfully uninstalled tokenizers-0.21.4
Attempting uninstall: openai
  Found existing installation: openai 2.43.0
  Uninstalling openai-2.43.0:
    Successfully uninstalled openai-2.43.0
Attempting uninstall: transformers
  Found existing installation: transformers 4.48.0
  Uninstalling transformers-4.48.0:
    Successfully uninstalled transformers-4.48.0
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
autotrain-advanced 0.8.36 requires huggingface-hub==0.27.0, but you
have huggingface-hub 0.36.2 which is incompatible.
autotrain-advanced 0.8.36 requires transformers==4.48.0, but you have
transformers 4.57.6 which is incompatible.
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3
which is incompatible.
```

google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.32.3 which is incompatible.
xarray-einstats 0.10.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
pytensor 2.38.3 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
tobler 0.14.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
opencv-python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
jaxlib 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
ydf 0.15.0 requires protobuf<7.0.0,>=5.29.1, but you have protobuf 4.25.9 which is incompatible.
grain 0.2.18 requires protobuf>=5.28.3, but you have protobuf 4.25.9 which is incompatible.
gradio 6.19.0 requires huggingface-hub<2.0,>=1.2.0, but you have huggingface-hub 0.36.2 which is incompatible.
gradio 6.19.0 requires starlette<2.0,>=1.0.1, but you have starlette 0.41.3 which is incompatible.
tiffiff 2026.4.11 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
gcsfs 2025.3.0 requires fsspec==2025.3.0, but you have fsspec 2024.9.0 which is incompatible.
opencv-contrib-python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
rasterio 1.5.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
shap 0.52.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
tensorflow 2.20.0 requires protobuf>=5.28.0, but you have protobuf 4.25.9 which is incompatible.
tensorflow 2.20.0 requires tensorboard~2.20.0, but you have tensorboard 2.18.0 which is incompatible.
grpcio-status 1.71.2 requires protobuf<6.0dev,>=5.26.1, but you have protobuf 4.25.9 which is incompatible.
jax 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
opencv-python-headless 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
Successfully installed aiohttp-3.9.5 gguf-0.19.0 huggingface-hub-0.36.2 numpy-1.26.4 openai-2.14.0 prometheus-client-0.20.0 protobuf-4.25.9 pytest-8.3.5 tokenizers-0.22.2 transformers-4.57.6 typer-0.24.2 wget-3.2

```
{"id": "13c71e1b28144395a6b2d0e708114da4", "pip_warning": {"packages": ["google", "numpy", "tokenizers", "transformers"]}}
```

Requirement already satisfied: huggingface_hub in
/usr/local/lib/python3.12/dist-packages (0.36.2)

Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub)
(3.29.4)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub)
(2024.9.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub) (1.5.1)
Requirement already satisfied: packaging>=20.9 in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub) (24.2)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub) (6.0.2)
Requirement already satisfied: requests in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub)
(2.32.3)
Requirement already satisfied: tqdm>=4.42.1 in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub)
(4.67.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub)
(4.15.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests-
>huggingface_hub) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests-
>huggingface_hub) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests-
>huggingface_hub) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests-
>huggingface_hub) (2026.6.17)

```
import os
import shutil
from huggingface_hub import snapshot_download

# Configuration
target_dir = "professional-assistant" # Le nom de votre projet
AutoTrain
base_model_id = "HuggingFaceTB/SmolLM2-360M-Instruct"
temp_repair = "repair_files"

print("📁 Téléchargement des fichiers de structure manquants...")
# On télécharge uniquement la config et le tokenizer originaux
snapshot_download(
    repo_id=base_model_id,
    allow_patterns=["*.json", "*.model", "*.txt"],
    local_dir=temp_repair
```

```

)

print(f"📁 Injection des fichiers dans {target_dir}...")
for file in os.listdir(temp_repair):
    src_path = os.path.join(temp_repair, file)
    dst_path = os.path.join(target_dir, file)

    # On ne copie que si le fichier n'existe pas déjà ou pour écraser
    les configs
    if os.path.isfile(src_path):
        shutil.copy(src_path, dst_path)
        print(f"📄 Ajouté : {file}")

print("\n📁 Dossier prêt pour la conversion !")

📁 Téléchargement des fichiers de structure manquants...

{"model_id": "191921eae8314b5e9615d983f721219d", "version_major": 2, "version_minor": 0}

{"model_id": "ba444c64f6b54ee9878b58a6c83a4c8d", "version_major": 2, "version_minor": 0}

{"model_id": "6a0ccf20aea149c6bb09ecb91d58f7a7", "version_major": 2, "version_minor": 0}

{"model_id": "fbca607a64ec466eb7d616da0c2c123d", "version_major": 2, "version_minor": 0}

{"model_id": "324b2b0e2b9a4b008c38a0fd21084768", "version_major": 2, "version_minor": 0}

📁 Injection des fichiers dans professional-assistant...
📄 Ajouté : eval_results.json
📄 Ajouté : all_results.json
📄 Ajouté : tokenizer.json
📄 Ajouté : merges.txt
📄 Ajouté : train_results.json
📄 Ajouté : tokenizer_config.json
📄 Ajouté : vocab.json
📄 Ajouté : config.json
📄 Ajouté : special_tokens_map.json
📄 Ajouté : generation_config.json
📄 Ajouté : trainer_state.json

📁 Dossier prêt pour la conversion !

# 3 : Conversion finale en GGUF

# Lancement de la conversion
# --outtype f16 : pour garder toute la précision du modèle
!python3 llama.cpp/convert_hf_to_gguf.py {target_dir}/ \

```

```
--outfile assistant_generique_final.gguf \  
--outtype f16
```

```
print("\n✅ SUCCÈS ! Votre modèle est prêt :  
'assistant_generique_final.gguf'")
```

```
INFO:hf-to-gguf:Loading model: professional-assistant  
INFO:hf-to-gguf:Model architecture: LlamaForCausalLM  
INFO:gguf.gguf_writer:gguf: This GGUF file is for Little Endian only  
INFO:hf-to-gguf:Exporting model...  
INFO:hf-to-gguf:Set meta model  
INFO:hf-to-gguf:Set model parameters  
INFO:hf-to-gguf:gguf: context length = 8192  
INFO:hf-to-gguf:gguf: embedding length = 960  
INFO:hf-to-gguf:gguf: feed forward length = 2560  
INFO:hf-to-gguf:gguf: head count = 15  
INFO:hf-to-gguf:gguf: key-value head count = 5  
INFO:hf-to-gguf:gguf: rope theta = 100000  
INFO:hf-to-gguf:gguf: rms norm epsilon = 1e-05  
INFO:hf-to-gguf:gguf: file type = 1  
INFO:hf-to-gguf:Set model quantization version  
INFO:hf-to-gguf:Set model tokenizer  
INFO:numexpr.utils:NumExpr defaulting to 2 threads.  
INFO:gguf.vocab:Adding 48900 merge(s).  
INFO:gguf.vocab:Setting special token type bos to 1  
INFO:gguf.vocab:Setting special token type eos to 2  
INFO:gguf.vocab:Setting special token type unk to 0  
INFO:gguf.vocab:Setting special token type pad to 2  
INFO:gguf.vocab:Setting chat_template to {% for message in messages %}  
{% if loop.first and messages[0]['role'] != 'system' %}{{ '<|im_start|  
>system  
You are a helpful AI assistant named SmolLM, trained by Hugging Face<|  
im_end|>  
' }}{% endif %}{{ '<|im_start|>' + message['role'] + '  
' + message['content'] + '<|im_end|>' + '  
' }}{% endfor %}{{ '% if add_generation_prompt %}{{ '<|im_start|>assistant  
' }}{% endif %}  
INFO:gguf.gguf_writer:Writing the following files:  
INFO:gguf.gguf_writer:assistant_generique_final.gguf: n_tensors = 0,  
total_size = negligible - metadata only  
Writing: 0.00byte [00:00, ?byte/s]  
INFO:hf-to-gguf:Model successfully exported to  
assistant_generique_final.gguf
```

```
✅ SUCCÈS ! Votre modèle est prêt : 'assistant_generique_final.gguf'
```

4 : Télécharger le modèle

```
from google.colab import files  
files.download('assistant_generique_final.gguf')
```



```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
```